

ANKUSH DESAI

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SUMMARY & ACHIEVEMENTS

Master's student in Robotics Technology with 10+ hands-on projects in robotic manipulation, autonomous navigation, and perception. Proficient in ROS/ROS2, Python, Computer Vision, Reinforcement Learning, Linear Control Systems, Mechanical Design, and Electronics. All India Rank 3 in ABU Robocon 2022, Rank 4 in 2021 and 2020. Seeking roles in Robotics, Perception, and Robotic Manipulation.

EDUCATION

University of Minnesota — Master of Science in Robotics Technology, GPA: 3.81/4

Sept. 2024 – Dec. 2026

Relevant Coursework: Robotics, Computer Vision, Machine Learning, Robot Vision, Linear Control Systems, Deep Learning, Optimal Estimation, Introduction to Nonlinear Optimization with ML & AI.

Pune University — Bachelor of Engineering in Mechanical Engineering, GPA: 3.4/4

Aug. 2019 – Jul. 2023

Relevant Coursework: CAD, Fluid Dynamics, Theory of Machines, Mechanical Design, Strength of Machinery, Thermodynamics.

EXPERIENCE

Pacific Northwest National Laboratory (PNNL) | Robotics Research Intern

Jul. 2026 – Nov. 2026

- Conducted frontier exploration to autonomously map an entire lab facility using an Orbbec RGB-D camera, generating dense 3D point clouds with ROS2, Nav2, and SLAM for real-time environment reconstruction and navigation.
- Applied Vision-Language-Action (VLA) models on a UR7e arm with a tactile-sensing Robotiq gripper for lab semantic understanding and chemical handling tasks (test-tube, well-plate handling), planning trajectories via MoveIt and integrating with autonomous navigation.
- Implemented TagSLAM for AprilTag-based localization to support GPS-denied lab navigation, and troubleshooted issues across the mapping, manipulation, and localization systems.

FAB Electronic Engineers | Intern, Automation Engineer

Jul. 2023 – Jul. 2024

- Developed an autonomous guided robot using ROS for navigation and obstacle avoidance, reducing paperwork delivery time by 90% with 92% accuracy and 98% efficiency through optimized path planning and sensor integration.
- Designed and manufactured a floor-cleaning robot with sweeping and mopping capabilities, reducing cleaning time by 50% through efficient control using a DPDT switchboard.

Team Automotons | Team Lead

Jun. 2022 – Jun. 2023

- Designed, manufactured, and tested two rugby-playing robots with 95% precision in control and coordination for ABU Robocon 2020, utilizing ROS, computer vision, and embedded systems.
- Applied a CNN-based system to classify hand movements using EEG data with 92% accuracy, enabling robotic arm control at 83% efficiency.

PROJECTS

[Videos of Projects](#)

Indoor Mapping Using Semantic 3D SLAM in ROS (ROS, Semantic Segmentation, Visual-SLAM, 3D Mapping)

- Built a semantic 3D SLAM algorithm for autonomous indoor navigation of TurtleBot 3 in GPS-denied environments, achieving 95%+ mapping/localization accuracy and a 70% improvement in navigation efficiency, validated in Isaac Sim.
- Enhanced the navigation system with dynamic obstacle avoidance and adaptive behavior modes, improving robustness across complex indoor scenarios.

Reinforcement Learning-based Robotic Manipulation with Koopman Generator (Reinforcement Learning, ROS)

- Built a PPO-based robotic picking system in NVIDIA Isaac Sim, achieving 30% faster task completion and 25% fewer collisions.
- Applied a Koopman-based linear model to simplify complex nonlinear robot dynamics, enabling smoother, more stable arm motion with a 40% reduction in erratic movements.

Autonomous Agricultural Rover for Weed Elimination (Python, ROS, Homogeneous Transformations, ML, NLP)

- Implemented a hybrid YOLOv7/VLM framework to identify 12 weed species with 95% accuracy for CO2-laser thermal ablation.
- Deployed a ROS-based path planning algorithm on the rover, achieving 92% collision avoidance during in-field navigation.

Autonomous Tool Sorting using a Computer Vision Algorithm (Computer Vision, ArUco markers, CNN, ML)

- Built a TensorFlow/Keras/OpenCV pipeline to control a UR5 robot for ArUco-marker-based tool identification and placement, achieving a 95% success rate in accurate bin placement.
- Extended the system to adapt to changes in tool position in real time, maintaining a 90% success rate under dynamic conditions.

SKILLS

- Python, PyTorch, C, C++, TensorFlow, MATLAB, Robot Operating System (ROS/ROS2), Nav2, MoveIt, Docker, Git, Linux, Claude Code.
- Computer Vision, Machine Learning, Deep Learning, Reinforcement Learning, Large Language Models (LLM), Vision-Language Models (VLM), Vision-Language-Action (VLA) Models, Robot Manipulation, SLAM, TagSLAM, AprilTag Localization, Path Planning, Linear Feedback Control.
- Robotic Arms (UR5, UR7e), Robotiq Grippers w/ Tactile Sensing, Orbbec RGB-D Cameras, Point Cloud Processing, Electrical Schematics, Electronic Debugging, Sensor Integration.
- 3D CAD (SolidWorks, Fusion 360, Catia), AutoCAD, ANSYS, Shapr 3D, Rapid Prototyping.